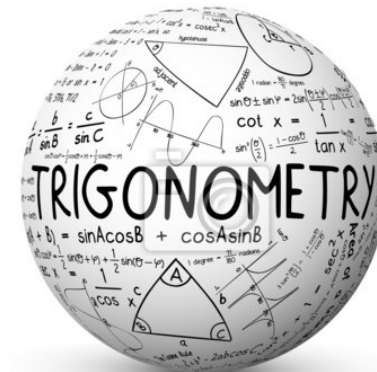
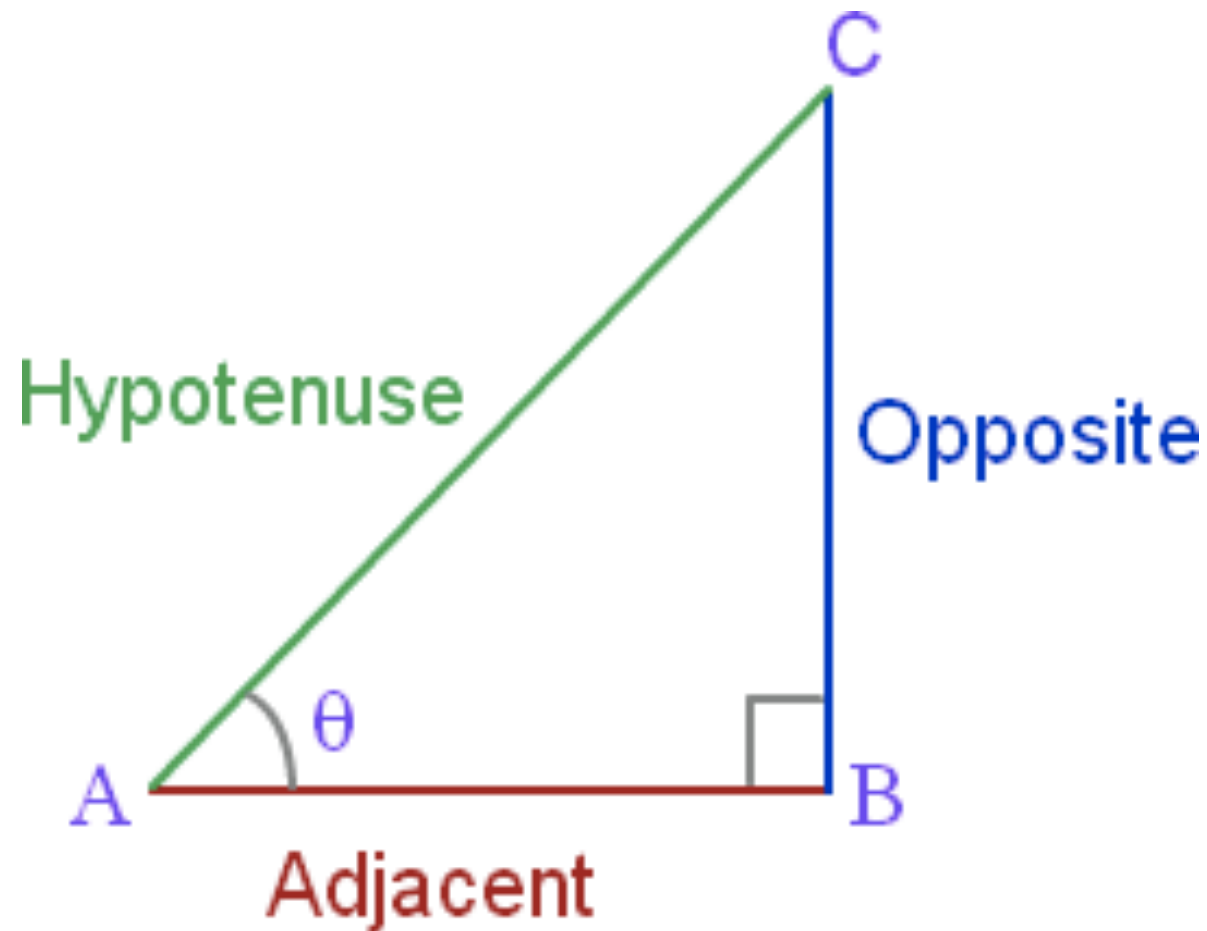




Trigonometric Ratios

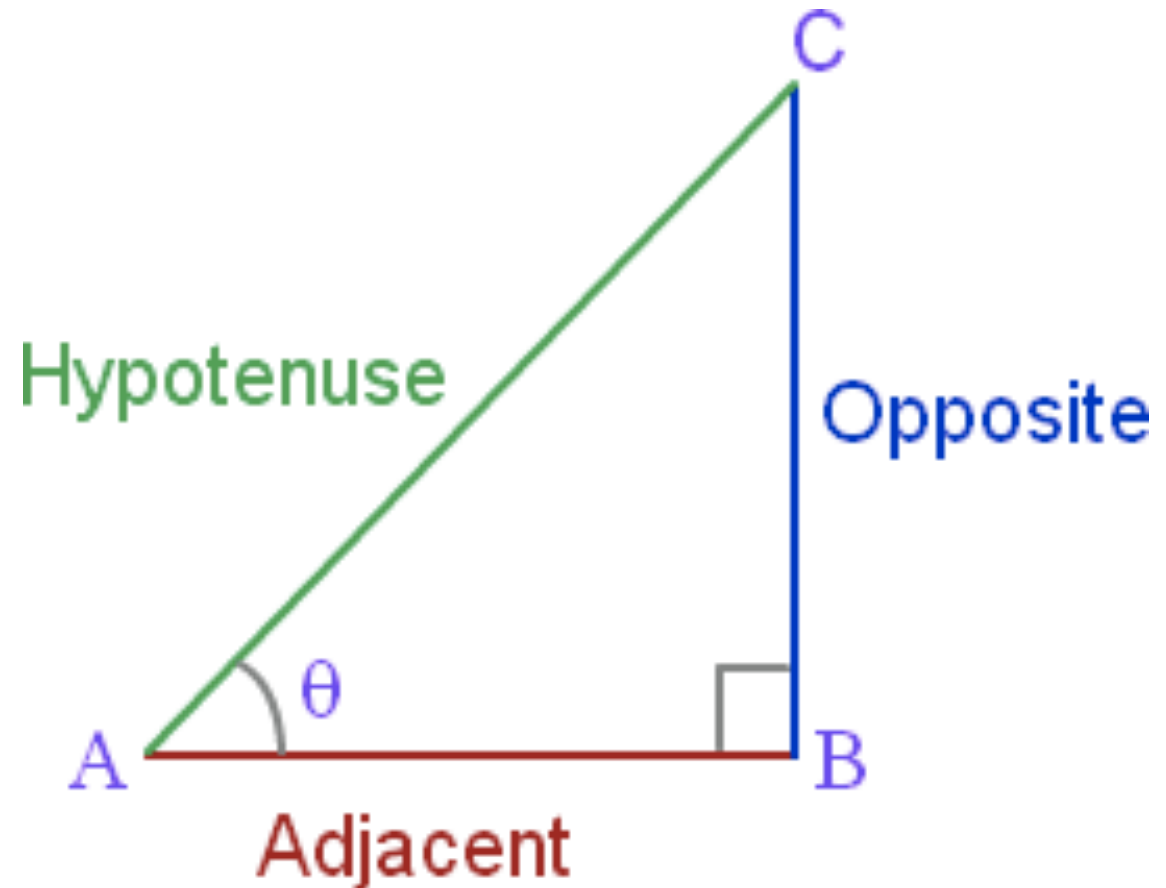


Sides of a Right Angled Triangle



Trigonometric ratios

- Sine (sin) = $\frac{\textit{Opposite}}{\textit{Hypotenuse}}$
- Cosine (cos) = $\frac{\textit{Adjacent}}{\textit{Hypotenuse}}$
- Tangent (tan) = $\frac{\textit{Opposite}}{\textit{Adjacent}}$



Sine	Cosine	Tangent
$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
SOH	CAH	TOA

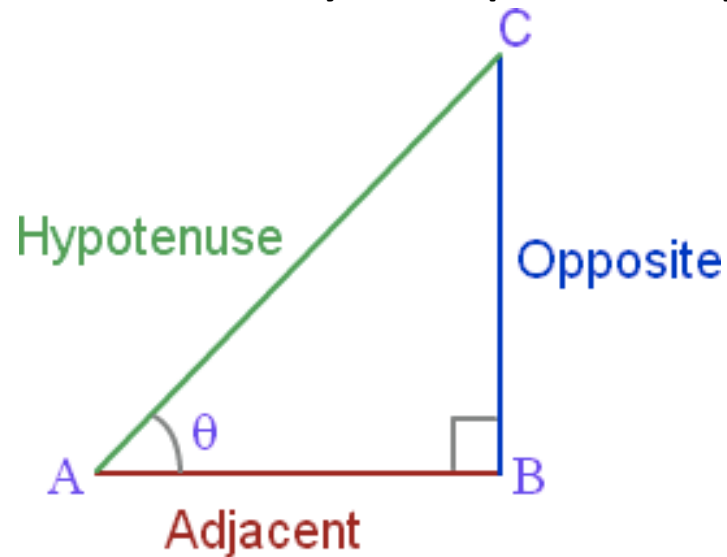
SOH CAH TOA

Some Old Hags Can Always Hide Their Old Age

Son Of Helen

Can Anyone Help

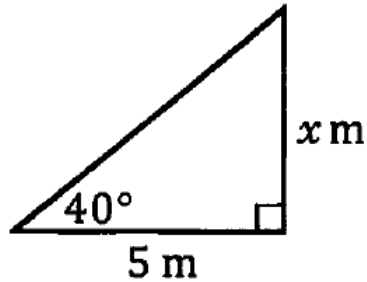
Try Once Again



Example 1

Find the value of x in each of the following, correct to one decimal place.

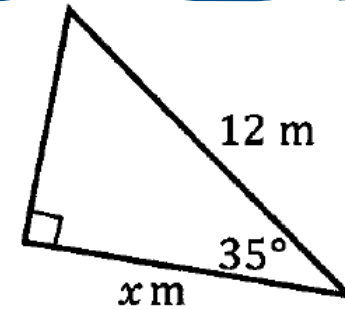
(a)



$$\tan 40^\circ = \frac{x}{5}$$

$$\begin{aligned} x &= 5 \tan 40^\circ \\ &= 4.195 \\ &\approx 4.2 \text{ m} \end{aligned}$$

(b)



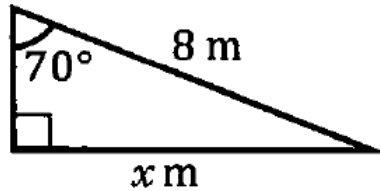
$$\cos 35^\circ = \frac{x}{12}$$

$$\begin{aligned} x &= 12 \cos 35^\circ \\ &= 9.83 \\ &\approx 9.8 \text{ m} \end{aligned}$$

Example 2

Find the value of x in each of the following, correct to one decimal place.

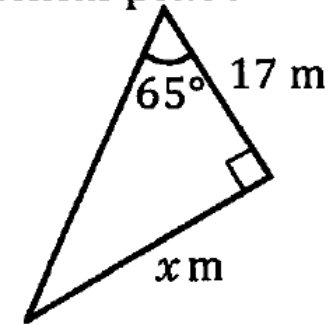
(a)



$$\sin 70^\circ = \frac{x}{8}$$

$$\begin{aligned} x &= 8 \sin 70^\circ \\ &= 7.5 \text{ m} \end{aligned}$$

(b)



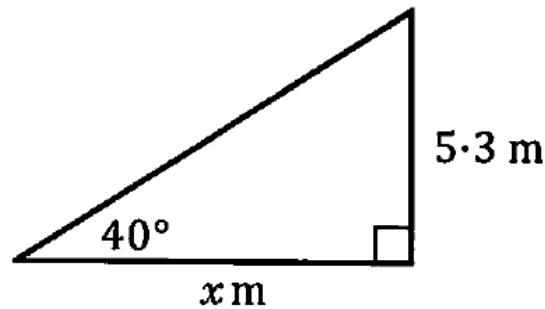
$$\tan 65^\circ = \frac{x}{17}$$

$$\begin{aligned} x &= 17 \tan 65^\circ \\ &= 36.5 \text{ m} \end{aligned}$$

Example 3

Find the value of x in each of the following, correct to one decimal place.

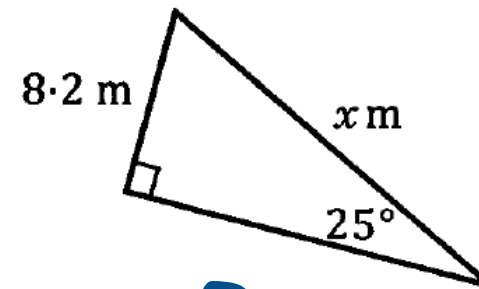
(a)



$$\tan 40^\circ = \frac{5.3}{x}$$

$$x = \frac{5.3}{\tan 40^\circ} \\ = 6.3 \text{ m}$$

(b)



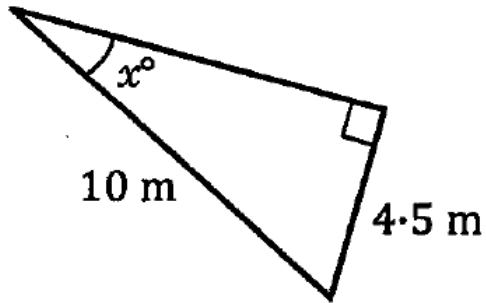
$$\sin 25^\circ = \frac{8.2}{x}$$

$$x = \frac{8.2}{\sin 25^\circ} \\ = 19.4 \text{ m}$$

Example 4

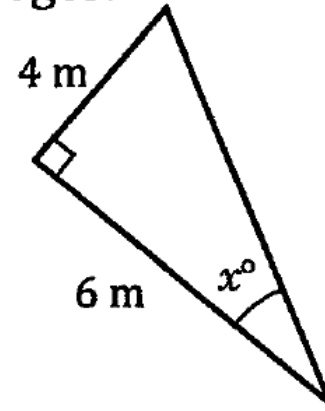
Find the value of x in each of the following, to the nearest integer.

(a)



$$\begin{aligned}\sin x^\circ &= \frac{4.5}{10} \\ x^\circ &= \sin^{-1}\left(\frac{4.5}{10}\right) \\ &= 26.74^\circ \\ &\approx 27^\circ\end{aligned}$$

(b)



$$\begin{aligned}\tan x^\circ &= \frac{4}{6} \\ x^\circ &= \tan^{-1}\left(\frac{4}{6}\right) \\ &= 33.69 \\ &\approx 34^\circ\end{aligned}$$

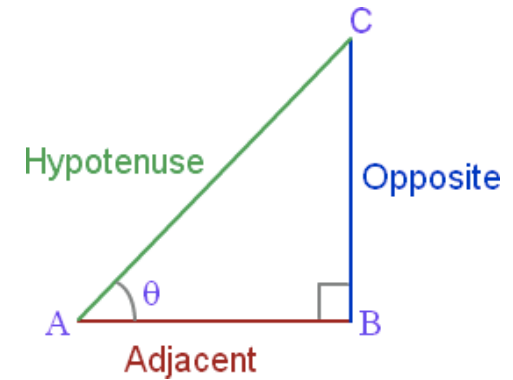
Given angle, θ

Determine the sides –
opposite, adjacent and
hypotenuse

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$



OPPOSITE
known

Find
ADJACENT

$$\text{Adjacent} = \frac{\text{Opposite}}{\tan \theta}$$

Find
HYPOTENUSE

$$\text{Hypotenuse} = \frac{\text{Opposite}}{\sin \theta}$$

ADJACENT
known

Find
OPPOSITE

$$\text{Opposite} = \text{Adjacent} \times \tan \theta$$

Find
HYPOTENUSE

$$\text{Hypotenuse} = \frac{\text{Adjacent}}{\cos \theta}$$

HYPOTENUSE
known

Find
ADJACENT

$$\text{Adjacent} = \text{Hypotenuse} \times \cos \theta$$

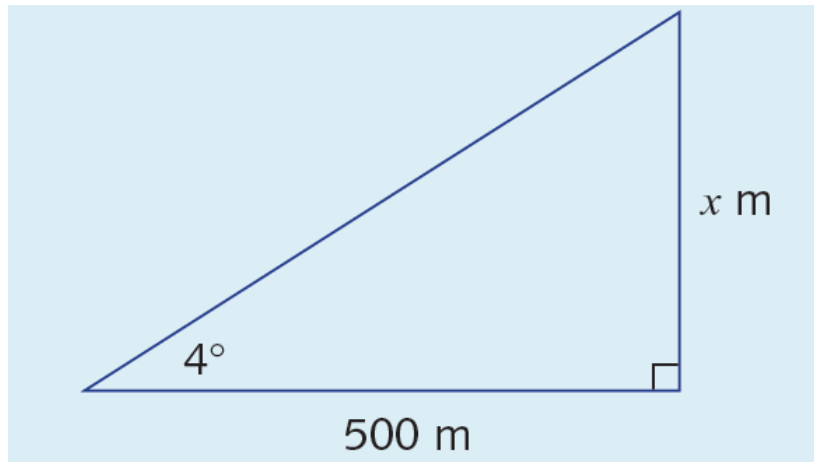
Find
OPPOSITE

$$\text{Opposite} = \text{Hypotenuse} \times \sin \theta$$

Finding Angles

- Use $\theta = \sin^{-1} \frac{O}{H}$ if you have Opposite and Hypotenuse
- Use $\theta = \cos^{-1} \frac{A}{H}$ if you have Adjacent and Hypotenuse
- Use $\theta = \tan^{-1} \frac{O}{A}$ if you have Opposite and Adjacent

A plane climbs at an angle of 4° after take-off. Find its altitude when it has flown a horizontal distance of 500 m.



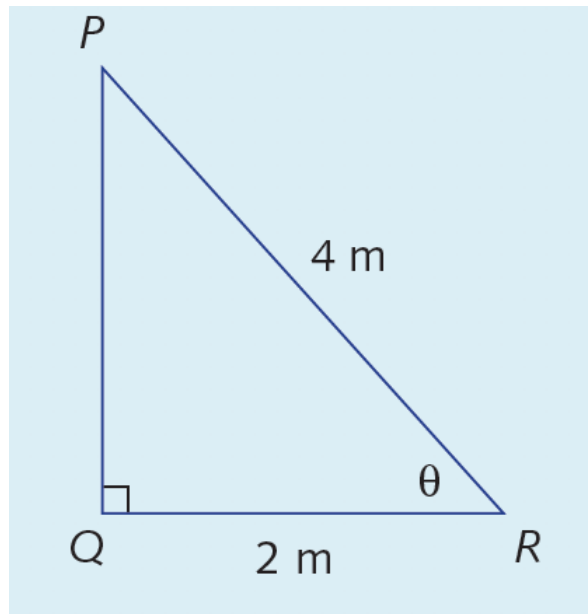
$$\tan 4^\circ = \frac{x}{500}$$

$$x = 500 \tan 4^\circ$$

$$= 34.96 \text{ m (2d.p.)}$$

A rope, 4 m long, is attached to a vertical pole. The rope, held taut, is pegged into the ground 2 m from the base of the pole.

- a Find the angle the rope makes with the ground.
- b Find the angle the rope makes with the pole.



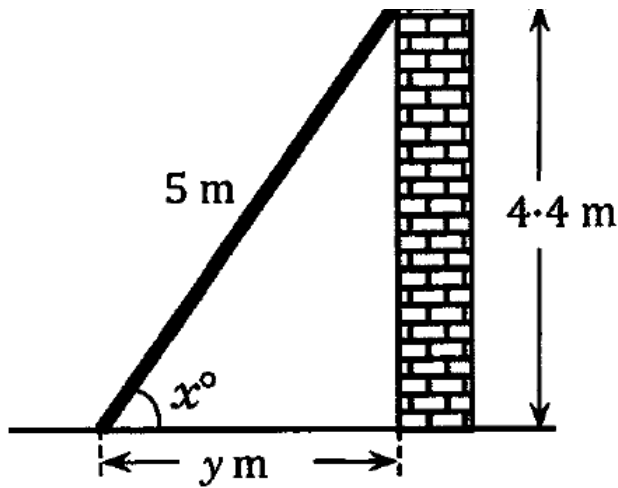
$$\begin{aligned} \text{a) } \cos \theta &= \frac{2}{4} \\ \theta &= \cos^{-1}\left(\frac{2}{4}\right) \\ &= 60^\circ \end{aligned}$$

The angle the rope makes with the ground is 60°

✓) The angle the rope makes with the pole is 30°

Example 5

A ladder of length 5 metres leans against a vertical wall and just reaches the top of the wall. If the wall is 4.4 metres high calculate the angle the ladder makes with the horizontal ground (to the nearest degree) and the distance from the foot of the ladder to the wall (in metres correct to one decimal place).



$$\sin x^\circ = \frac{4.4}{5}$$

$$x^\circ = \sin^{-1}\left(\frac{4.4}{5}\right)$$

$$= 61.64$$

$$\approx 62^\circ$$

Do NOT use 62° for

$$\cos x^\circ = \frac{y}{5}$$

$$y = 5 \cos x^\circ$$

$$= 2.37$$

$$\approx 2.4 \text{ m}$$